Base paper: architecture as a block diagram

Subramanian, Syeda-Mahmood and Do, "Modelling-based joint embedding of histology and genomics using canonical correlation analysis for breast cancer survival prediction," *Artificial Intelligence in Medicine* 149:102787, 2024. Built from the paper and the authors' released code (01_methods, 03_brca).

TCGA-BRCA cohort

974 patients with both modalities and survival (time T, event δ). 95 deaths (9.8%), 90% censored.

Genomics branch

RNA-seq (FireBrowse) → log2 expression → top 1000 genes by coefficient of variation → z-scores

$X \in \mathbb{R}^{1000}$

Histology branch

H&E whole-slide images → nuclei segmentation → 25 patches (2000×2000) per patient → CellProfiler, 215 features → 5-bin histograms

$Y \in \mathbb{R}^{1075}$

Splits and normalisation

5 fixed folds: 585 train / 146 valid / 243 test. Features normalised with training statistics.

→

STAGE 1 · UNSUPERVISED EMBEDDING (penalised CCA, "pCCA")

Find one pair of directions

$\max_{u,v} \text{corr}(u^T X, v^T Y)$

subject to a penalty on u, v

Three variants

CCA: no penalty (overfits when p ≫ n)
SCCA: L1 sparsity, few features per direction
GN-SCCA: L1 + GraphNet penalty, graph from feature covariance, related features picked together

Hyperparameters

Grid over penalty weights; best set = highest canonical correlation on a held-out split.

Deflation, repeat j = 1…K

Remove what (u_j, v_j) explained, find the next pair.
HD: Hotelling · PD: projection · OPD: orthogonalised projection

⬇

Learned projections (K = 50)

$U = [u_1 \ldots u_K]$ 1000 × 50

$V = [v_1 \ldots v_K]$ 1075 × 50

⬇

Patient embedding

Genomics $U^T x$ · Imaging $V^T y$

Concatenated $[U^T x, V^T y]$ (100-d)

→

STAGE 2 · LABEL PREDICTION

Cox proportional hazards

Fitted on training embeddings (lifelines, penalizer 0.1)

$h(t|e) = h_0(t)\, exp(\beta^T e)$

⬇

Evaluation

Risk score $\beta^T e$ on test patients → Harrell's C-index, mean ± sd over 5 folds

Baseline

PCA: 50 genomic + 50 imaging PCs → same Cox model

Sanity check: simulations

Data generated from a latent-variable model (z → x, z → y; d = 5, p = q = 200, n = 100) to test whether each method recovers the shared latent factor

Key result (concatenated, C-index %)	PCA	SCCA / HD	SCCA / PD	SCCA / OPD (best)	GN-SCCA / PD	GN-SCCA / OPD	GN-SCCA / HD
Authors' stored results, reproduced exactly by us	66.47 ± 6.86	67.79 ± 8.26	67.58 ± 4.69	**68.98 ± 6.16**	65.47 ± 10.14	66.85 ± 9.47	failed

Our architecture and planned innovation

Same data, folds, Cox model and C-index as the paper (fair comparison); Stage 1 becomes an explicit PGM. same as paper · new, done in M2 · planned, M3

same

Inputs

TCGA-BRCA, same 5 folds, train-only standardisation.
M2: paper's PCA input (50 genomic + 50 imaging PCs).
M3: full 1000 genes + 1075 histology features.

same

Paper baseline

Authors' trained SCCA / GN-SCCA models and survival code, rerun: 0.00 difference.

→

STAGE 1 · PROBABILISTIC GRAPHICAL MODEL (replaces pCCA + deflation)

new

Representation: probabilistic CCA

$z \sim N(0, I_d)$

$x \mid z \sim N(W_x z + \mu_x, \Psi_x)$

$y \mid z \sim N(W_y z + \mu_y, \Psi_y)$

Graph z → x, z → y, so **X ⊥ Y | Z** (block-diagonal Ψ)

new

Learning (maximum likelihood)

CCA-MLE: closed-form solution (baseline route)
EM: E-step posterior of z, M-step updates W, Ψ (our extension)
d chosen on validation C-index

new

Inference: exact Gaussian posterior

$p(z \mid x, y)$ both modalities
$p(z \mid x)$ genomics only · $p(z \mid y)$ histology only
MAP = posterior mean ẑ; uncertainty = posterior entropy

⬇

M3

Missing modality

Patient lacks x or y: drop that term from the posterior (exact marginalisation), compared with imputation. The paper's pipeline needs both inputs.

⬇

Patient embedding

ẑ = E[z | observed data] (d-dim), plus its uncertainty

→

STAGE 2 · PREDICTION

same

Cox PH on ẑ

L2 penalty 0.1, risk = $\beta^T \hat{z}$

⬇

same

C-index

mean ± sd over 5 folds

M2 result (paper PCA input)

CCA-MLE 65.43 ± 6.58
EM 63.44 ± 8.09
Paper best 68.98 ± 6.16
Competitive, within one sd, not better yet.

#	Innovation over the base paper	What it adds	Status
1	Stage 1 as an explicit PGM (probabilistic CCA)	The paper motivates CCA by a latent-variable model but only optimises correlation; we fit that model's probabilities and use its conditional-independence structure.	Done (M2)
2	EM parameter learning	A second, iterative maximum-likelihood route alongside the closed form; works with missing data and extends to priors.	Done (M2)
3	Probabilistic embedding with uncertainty	Each patient gets a posterior, not a fixed projection; entropy shows how much each modality informs z.	Done (M2)
4	Exact missing-modality prediction	Predict survival when histology or genomics is unavailable, by marginalisation instead of imputation. Main contribution.	M3
5	Combine the paper's strengths with the PGM	EM started from the authors' SCCA weights; sparsity through a prior on W (MAP-EM); full 1000 / 1075 features with diagonal or low-rank noise.	M3